

ForenSmart Expert Diagnostic Report

ForenSmart Project - Consent ID Persistence Problem

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Environment: Streamlit + Python 3.11 (ForenSmart App)

Primary File: app_patched_fixed_final.py

1. Overview

The application intermittently fails to detect the recorded consent ID between runs or between modules (main UI and adapter processes). Although a consent record (consent_records/<id>.json) is successfully written to disk, the adapters occasionally display:

"No consent_id linked in session. Capture consent in the Consent area before running intrusive extractions."

This occurs even when a valid consent file exists and has been verified manually.

2. Implemented and Verified

- Streamlit session handling: Implemented, session_state['consent_id'] is set during consent capture
- File persistence: Working, JSON file saved under consent_records/ with valid content and correct timestamp
- Atomic write helper: Added, uses temporary file and atomic rename to avoid partial writes
- Session restore helper: Added, restore_consent_into_session(st) executes after imports
- Consent file verification: Confirmed, JSON is valid and readable
- Adapter layer check: Fails intermittently; adapters sometimes cannot see consent_id

3. Symptoms

- Consent recorded successfully in UI; consent_records/<id>.json is created
- After refreshing Streamlit or launching adapters:
- Session sometimes lacks consent_id

- Manual file read shows consent exists, indicating a restore or visibility issue

4. Possible Root Causes

1. Multi-process isolation
2. Working-directory mismatch
3. Race condition (timing between write and read)
4. Permissions or environment differences
5. Multiple sessions or browser tabs with empty session_state

5. Diagnostic Evidence

- Consent JSON exists after save
- session_state contains consent_id immediately after recording
- After reload, session_state may be empty
- Manual file read returns correct consent id
- Adapter warning repeats until restart

6. Temporary Fixes Attempted

- Consent bypass: Works for testing
- Auto-restore on startup: Reduces frequency, not perfect
- Atomic JSON write: Prevents malformed files
- Reordering restore call: Minimal improvement

7. Next Expert Investigations

- Process topology review (subprocesses, working directory)
- Environment audit (Docker/VM, mounts, permissions)
- Session-restore logging (trace restore timing)
- Centralized consent service (database or API)
- Absolute file paths and normalization
- Check Streamlit reload behavior (--server.runOnSave flag)

8. Recommended Expert Deliverables

- Process diagram showing Streamlit and adapter interactions
- Verification test for shared visibility of consent_records
- Proposal for shared persistence (DB, KV-store, or IPC)

9. Files and Logs for Review

- app_patched_fixed_prod.py
- consent_records/ directory samples
- ruff_report.txt
- Streamlit runtime logs
- Optional: debug_session_output.txt

10. Summary

Consent ID logic functions inside the Streamlit process but intermittently fails between processes due to isolation or visibility issues. All internal code defects fixed. Expert review needed for runtime topology and persistence layer design.